

The isotropy-infimum core of Math427: angular-momentum block-diagonal

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

Since SC-SCOPE was lifted (2026-06-09), B1-RH-ENUM rests on the SOLE hypothesis H-LAYER, whose analytic core is Math427: within the diagonal-Gaussian class the isotropic dressing is the variational infimum. The companion note (hdiag-fullcovariance-formulation v1.0) proved the condensate-free functional F_0 convex; T-018 (hdiag-offdiag-gershgorin-threshold v1.1) reduced the residual off-diagonal obstruction to a single exchange scalar. This note completes the DIAGONAL half: it makes the isotropy of the infimum a STRICT, curvature-protected statement, exhibiting the exact mechanism by which rotation invariance forces the minimiser to be isotropic. Scope: F_0 over the diagonal-Gaussian class (H-diag), second cumulant, anchor $\mu^2 = 5 \times 10^{-3}$.

2. The condensate-free functional and its second variation

For a diagonal homogeneous Gaussian trial $G(q)$,

$$\frac{F_0[G]}{V} = \frac{1}{2} \int_q \ln G(q)^{-1} + \frac{1}{2} \int_q K_0(q) G(q) + \Phi_{\text{diag}}(M_{\text{tot}}), \quad K_0(q) = c(q^2 - q_0^2)^2, \quad M_{\text{tot}} = \int_q G(q),$$

with $\Phi_{\text{diag}}(M) = \frac{3u}{4} M^2 + \frac{5v}{2} M^3$ (Math427). The interaction depends on the trial state ONLY through the rotation-invariant scalar M_{tot} . The second variation at the isotropic dressing $G_*(q) = 1/(r_R + c(q^2 - q_0^2)^2)$ is

$$\delta^2 F_0 = \frac{1}{2} \int_q G_*(q)^{-2} \delta G(q)^2 + \Phi''_{\text{diag}}(M_{\text{tot}}) (\delta M_{\text{tot}})^2, \quad \delta M_{\text{tot}} = \int_q \delta G(q),$$

entropy contributing $\frac{1}{2} G_*^{-2} > 0$ per mode, the kinetic term linear (no curvature), the interaction through δM_{tot} only.

3. Angular-momentum block-diagonalisation (the mechanism)

Expand $\delta G(q) = \sum_{l,m} a_{lm}(|q|) Y_{lm}(\hat{q})$. Because $\int Y_{lm} d\Omega = \sqrt{4\pi} \delta_{l0} \delta_{m0}$, ONLY the $l = 0$ component survives δM_{tot} . By orthonormality of $\{Y_{lm}\}$ the entropy quadratic form is already diagonal in (l, m) , so the full Hessian is BLOCK-DIAGONAL in angular momentum. Writing the $l = 0$ radial mode $\delta G_{00}(q)$ with radial measure $d\mu(q) = |q|^2 d|q| / (2\pi)^3$, the interaction term is the explicit RANK-ONE positive form

$$\delta^2 F_0[\delta G_{00}] = \frac{1}{2} \int_0^\infty G_*(q)^{-2} |\delta G_{00}(q)|^2 d\mu(q) + \Phi''_{\text{diag}}(M_R) \left(\int_0^\infty \delta G_{00}(q) d\mu(q) \right)^2, \quad H_{l \geq 1} = \frac{1}{2} G_*^{-2}$$

i.e. the rotation-invariant interaction curves ONLY the isotropic breathing sector $l = 0$ (a rank-one positive operator built from the M_{tot} functional); every anisotropic sector $l \geq 1$ is interaction-FLAT.

4. Per-sector positivity (strict minimum, analytic)

(a) $l = 0$ (**breathing**). $\Phi''_{\text{diag}}(M) = \frac{3}{2}(u + 10vM) = \frac{3}{2}u_{\text{eff}}$; at the anchor $u_{\text{eff}} = -0.86 + 10(3.24)(0.10941) = +2.685$, so $\Phi''_{\text{diag}} = +4.03 > 0$: the rank-one interaction form is positive semidefinite, and the breathing sector retains in addition the strictly positive entropy term. (b) $l \geq 1$ (**anisotropic**). $H_{l \geq 1} = \frac{1}{2}G_*^{-2}$, with the ANALYTIC infimum

$$\frac{1}{2}G_*(q)^{-2} = \frac{1}{2}[r_R + c(q^2 - q_0^2)^2]^2 \geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0,$$

attained at the gap shell $|q| = q_0$. Positivity is therefore ANALYTIC (a square plus r_R^2), not a numerical grid value; the grid minimum equals $\frac{1}{2}r_R^2$ to $< 10^{-3}$. Both sectors are strictly positive, so G_* is a STRICT minimum of F_0 within the diagonal class: isotropy is protected by a positive curvature gap in EVERY angular-momentum sector. The mechanism is verified by exact Gauss–Legendre quadrature ($\int Y_{lm} d\Omega = 0$ for $l \in \{1, 2, 3, 4\}$ to 1.8×10^{-15}).

5. What remains (the honest residual)

This certifies the DIAGONAL half of Math427 (conditional on H-diag) as a strict, curvature-protected minimum, with analytic per-sector positivity. Two residuals remain, both mapped: (i) H-diag – the restriction to diagonal covariance – is Math427’s standing hypothesis; (ii) the condensate-induced OFF-diagonal Hessian, which hdiag-fullcovariance-formulation isolated and T-018 reduced to the single competitor-dependent exchange scalar $b_{\text{exch}}(Q) < b_*(Q)$ (binding full-class target 0.0206; the dressed Fock bubble, RES-5). Thus H-LAYER’s analytic core now reads: the diagonal isotropy infimum is a STRICT minimum (this note, analytic), and the only obstruction to the full covariance is the exchange scalar (RES-5). No tier flip: B2 T6, B1 T6 on {H-LAYER}.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “The $l = 0$ interaction term was written as a scalar product $\Phi''_{\text{diag}} w$.” ADDRESSED (operator review): §3 now writes it as the explicit RANK-ONE positive form $\Phi''_{\text{diag}}(M_R)(\int \delta G_{00} d\mu)^2$ built from the M_{tot} functional; positive semidefinite since $\Phi''_{\text{diag}} > 0$.
- (β) “The entropy floor is numerical.” RESOLVED (operator review): $\frac{1}{2}G_*^{-2} = \frac{1}{2}[r_R + c(q^2 - q_0^2)^2]^2 \geq \frac{1}{2}r_R^2 > 0$ is analytic, attained at $|q| = q_0$; the grid value 4.64×10^{-2} IS $\frac{1}{2}r_R^2$. The $l \geq 1$ protection is theorem-grade. (Headline tier kept T4 because the DIAGONAL restriction H-diag and the off-diagonal residual remain.)
- (γ) “Does this discharge H-LAYER or flip B1?” DISMISSED: it strengthens the DIAGONAL core (conditional on H-diag) and leaves the off-diagonal exchange scalar (RES-5) as the residual. B2 stays T6, B1 stays T6 on {H-LAYER}.

Quantitative sanity check (mandatory): *mechanism* – Φ_{diag} depends only on rotation-invariant M_{tot} , Hessian on $l = 0$ alone ($\int Y_{l \geq 1} d\Omega = 1.8 \times 10^{-15}$); *sign* – bare $u = -0.86 < 0$ lifted to $u_{\text{eff}} = +2.685 > 0$, $\Phi''_{\text{diag}} = +4.03 > 0$; *entropy* – analytic $\geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$; *consistency* – grid min = $\frac{1}{2}r_R^2$.

7. Result footer

Result ID:	B2-PROPA-HLAYER / Math427 isotropy-infimum core (v1.1 rank-one + analytic-infimum re-issue)
Precise statement:	Condensate-free F_0 over the diagonal-Gaussian class. The second variation at G_* is $\delta^2 F_0 = (1/2) \int G_*^{-2} (dG)^2 + \Phi''_{\text{diag}} (dM_{\text{tot}})^2$. Since Φ_{diag} depends only on rotation-invariant $M_{\text{tot}} = \int G$ and $\int Y_{lm} d\Omega = 0$ for $l > 1$, the Hessian is BLOCK-DIAGONAL in angular momentum: $H_{\{l=0\}} = (1/2) \int G_*^{-2} dG_{00} ^2 d\mu + \Phi''_{\text{diag}}(M_R) (\int dG_{00} d\mu)^2$ [rank-one positive]; $H_{\{l>1\}} = (1/2) G_*^{-2} \geq (1/2) r_R^{-2}$ [analytic, attained at $ q =q_0$]. Both strictly positive ($\Phi''_{\text{diag}} = 4.03 > 0$; $(1/2) r_R^{-2} = 4.64e-2 > 0$) $\Rightarrow G_*$ is a STRICT diagonal minimum; isotropy is curvature-protected per sector, not fine-tuned.
Scope:	Diagonal-Gaussian class (H-diag); second cumulant; anchor $\mu^2 = 0.005$. Off-diagonal extension = T-018 exchange scalar.
Dependencies:	Math427 (diagonal isotropy theorem); hdiag-fullcovariance-formulation (F_0 convexity, u_{eff}); hdiag-offdiag-gershgorin-threshold v1.1 (T-018 off-diagonal residual); Math424-AddA.
Evidence grade:	ANALYTIC (exact block-diagonalisation + analytic entropy infimum $(1/2) r_R^{-2}$) + EXECUTED (res5_016_isotropy_infimum_core.py v1.1 6/6, GL exact quadrature).
Reproduction command:	python codes/vacuum/res5_016_isotropy_infimum_core.py
Expected output:	$\Phi''_{\text{diag}} = 4.03 > 0$; entropy infimum $(1/2) r_R^{-2} = 4.64e-2 > 0$; $\langle Y_{\{l>1\}} \rangle \sim 2e-15$; 6/6 PASS.
Falsification gate:	An anisotropic ($l > 1$) diagonal perturbation that lowers F_0 (would require the interaction to curve $l > 1$, contradicting rotation invariance of M_{tot}); or $\Phi''_{\text{diag}} \leq 0$ ($u_{\text{eff}} \leq 0$, refuted: $u_{\text{eff}} = +2.685$).
Tier before / after:	No tier flip. B2 T6, B1 T6 on {H-LAYER}. Math427 diagonal core: strengthened from stationary-infimum to STRICT curvature-protected minimum, with ANALYTIC per-sector positivity (theorem-grade within the diagonal class).
No-overclaim statement:	Certifies only the DIAGONAL half (conditional on H-diag) as a strict minimum. Does NOT discharge H-LAYER and does NOT flip B1/B2. The off-diagonal residual is the T-018 exchange scalar $b_{\text{exch}}(Q) < b_{\text{star}}(Q)$ (binding 0.0206, RES-5). H-diag remains Math427's standing hypothesis.
Next required action:	Bound the exchange scalar $b_{\text{exch}}(Q)$ at operating intensity (RES-5/GAP-2) -- the single residual that, with this note and T-018, would extend the isotropy infimum from the diagonal class to the full covariance and discharge H-diag.